

Gender bias in team formation: the case of the European Science Foundation's grants

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This paper investigates gender bias (if any) when teams are formed. We use data from the European Science Foundation to estimate if female scientists have the same opportunities as their male colleagues to join a team when applying for funds. To assess gender bias, we construct a control group of scientists with the competencies for being invited to join the team but who do not join. By comparing the proportion of female scientists in the control group with the one in the observed teams, we find a gender bias against female scientists. Exploring heterogeneity across teams, we find that gender bias is less pronounced in teams led by women, junior scientists, and in more recently formed teams. We also observe differences across disciplines.

Keywords: gender bias; team science; team selection; research grants.

1. Introduction

Despite the call for gender parity coming from private and public actors, policymakers, and also advocated among the Sustainable Development Goals by the United Nations General Assembly, a remarkably significant gender gap against women persists across different sectors of our society. Academia is not exempt from gender inequality: female scientists suffer from lower productivity (Lee and Bozeman 2005; Arensbergen, van der Weijden, and Besselaar 2012; Kelchtermans and Veugelers 2013; Pezzoni et al. 2016; Wagner 2016; Lerback, Hanson, and Wooden 2020; Ross et al. 2022), fewer promotion opportunities (McDowell, Singell, and Ziliak 1999; Ceci and Williams 2011; Marini and Meschitti 2018; Meschitti and Marini 2022), and lower probability of obtaining funds than their male colleagues (Lawson, Geuna, and Finardi 2021). Among the factors that help women to close their productivity gap, previous studies have shown the benefit of working in teams (Fox and Mohapatra 2007; Badar, Hite, and Badir 2013; Fox 2020).

It is well recognized that scientists benefit from working in teams (Jones 2021), and it becomes crucial to understand if female and male scientists have equal opportunities to enter a team. Extant studies have considered the propensity of individual scientists to collaborate (Abramo, D'Angelo, and Di Costa 2009; Abramo, D'Angelo, and Murgia 2017), while a few have documented heterogeneity in collaboration behaviors across genders (Boschini and Sjögren 2007; Zeng et al. 2016). However, no studies have considered whether female

scientists have the same probability as their male colleagues of being selected as team members when a team is formed. Indeed, scientists do not randomly join a team. They team up according to their characteristics, background, specialization, and skills. Therefore, this paper aims to answer the question: does gender affect team formation?

We rely on a unique dataset of 329 teams of scientists applying to a European Science Foundation (ESF)'s granting program called EUROCORES between 2006 and 2010. The ESF is a nongovernmental nonprofit foundation established in France in 1974 to promote high-quality research and innovation. The program EUROCORES was launched to promote an 'independent, bottom-up approach to collaborative research in Europe that allowed new scientific ideas to be developed' (ESF 2015) and required one project leader (PL) to team up with other principal investigators (PIs) to respond to an open call. To assess the gender bias in team formation (if any), we needed to assess the presence of female scientists in a pool of potential PI candidates that the PL might have invited to join the team, but the PL did not invite. Being limited by observing only the final composition of the 329 teams analyzed, we implemented a quasi-natural experiment constructing a control group of potential teammates. Specifically, we compared the probability of having a female scientist in a control group of scientists with the potential to be selected with the probability of having a female scientist in the observed teams. Potential PIs are selected assuming three main mechanisms used by PLs to screen PIs. First, we assume that a PL invites PIs to join

the team by searching among the scientists with whom she has already collaborated. Then, a PL looks for PIs among colleagues she knows for their research work. Finally, a PL looks for PIs in university departments working in the research areas of interest for the project application. Once we identified all the eligible controls in these three pools of scientists, we paired each observed PI with a corresponding potential PI with the most similar characteristics. Finally, we compared the probability of observing a female PI in the control group and the observed teams.

We find a gender bias in the team formation phase. The probability of having a female PI in an observed team equals only 17.5 per cent, compared to the 22.6 per cent probability of having a female scientist in the control group of potential PIs. Digging into this result, we observe that gender bias is less pronounced in teams led by women, junior scientist PLs, and teams applying for the grant in more recent years. Similarly, when looking at the disciplines, physics and engineering are gender-neutral, while life and medical sciences, and humanities are gender biased against women. Although gender bias against women is observed for high- and low-productive PLs, the former show a less pronounced bias. We also find gender bias in teams led by PLs working in both high and low-masculine countries. However, gender bias is more pronounced for those working in high masculine countries. Finally, the project quality, as represented by being awarded the grant, does not affect gender bias in PIs' selection: neither awarded nor not awarded teams are gender biased.

Our work makes three main contributions. First, we add to the literature on team composition. The extant contributions focus on the role played by women in research teams (Sosik and Godshalk 2000; Ivanova-Stenzel and Kübler 2005; Bandiera, Barankay, and Rasul 2013; West *et al.* 2013; Dasgupta, Scircle, and Hunsinger 2015; Pezzoni *et al.* 2016; Gaulé and Piacentini 2018). Although those studies have contributed to assessing how gender composition makes teams more productive, they have neglected that the observed teams' composition results from a selection process of individuals. We add to this literature by investigating the selection phase. Investigating selection is crucial because, as the previous literature shows, team gender composition influences the team's performance. For instance, a gender bias against women might lead to not selecting women with competencies and skills that would have fitted the project's aims, causing a decrease in team performance. Our second contribution is that our empirical setting allows consider successful and unsuccessful teams. When reconstructing teams of scientists, extant studies look at coauthorships in published papers (Boschini and Sjögren 2007; Abramo, D'Angelo, and Di Costa 2009; Abramo, D'Angelo, and Murgia 2017). By doing so, they observe teams that successfully deliver publication outcomes. In our work, we consider teams formed to compete in fundraising, regardless of the result of the competition. Our choice of the empirical setting mitigates a potential bias resulting from observing only successful teams of superior quality. Finally, we add to the gender parity debate. Several statistics report that women are underrepresented in research teams (Abramo, D'Angelo, and Di Costa 2009). However, a lower number of women in a team do not necessarily denote a gender bias against women because it might result from a scarcity of women among the potential teammates to select (Card *et al.* 2023). By comparing the proportion of potentially qualified

women to enter a team with those who entered teams, our findings confirm gender bias against women in the selection, even if the teams' characteristics affect the bias extent.

The paper is structured as follows. Section 2 reviews the literature on teamwork in science. Section 3 describes our empirical strategy to assess the existence of a possible gender bias in team formation. Section 4 presents our data and describes how we construct the control group of potential PL's teammates. Section 5 shows the results, and Section 6 concludes.

2. Teamwork in science

Teamwork is a growing phenomenon. Within the scientific community, Wuchty, Jones, and Uzzi (2007) analyzed paper coauthorships and patent co-inventorships between the late 50s and 2000 over a sample of about 20 million papers and 2 million patents, finding that teamwork has become the dominant working mode to produce knowledge in all fields. Not only individuals prefer teamwork to solo work, but when joining their efforts, their outcomes have a higher impact (Wuchty, Jones, and Uzzi 2007; Uzzi *et al.* 2013; Wagner, Whetsell, and Leydesdorff 2017). For example, multiple authors' papers receive more citations than solo-author ones (Wuchty, Jones, and Uzzi 2007; Freeman and Huang 2014). Teamwork also favors creativity (Jones 2021). Recently, Wu, Wang, and Evans (2019) examined more than 65 million papers, patents, and software products realized over 60 years, from 1954 to 2014, and found that teams play a crucial role in the science and technology landscape. In particular, small teams produce disruptive outcomes with high impact.

Individuals team up to complement their knowledge and skills in response to the so-called 'burden of knowledge' challenge (Jones 2009, 2021). For single individuals, it is impossible to acquire all the scientific knowledge accumulated in their disciplines over decades, so they specialize in narrow areas of expertise. At the same time, the complexity of the problems tackled by modern science often requires interdisciplinary contributors (Falk-Krzesinski *et al.* 2011; Milojević 2014). As a consequence, individuals team up to complete complex tasks. By surveying almost 4,000 corresponding authors involved in collaborations with at least one US coauthor in the fields of particle and field physics, nanoscience and nanotechnology, and biology and applied microbiology, Freeman, Ganguli, and Murciano-Goroff (2014) find that searching for unique human capital is the main driver for collaborations: scientists opt for the division of labor and look for 'unique knowledge, expertise, and capabilities' (14). Finally, another reason for individuals to team up is to access special equipment or infrastructures that would not be accessible to single individuals due to their high cost (Stephan 1996; Katz and Martin 1997).

If teamwork allows individuals to access complementary skills and unique resources to boost productivity, creativity, and impact, it becomes crucial to understand how individuals access a team. In recent years, collaboration costs have decreased, and teamwork faces lower entry barriers. The broad-scale adoption of the Internet has facilitated long-distance communication (Cairncross 1997), and the substantial reduction of traveling costs made possible through solutions like access to low-cost flight connections boosted

collaborations. For instance, [Catalini, Fons-Rosen, and Gaulé \(2020\)](#) analyzed the effect of the entry of the low-cost Southwest in the airlines market, finding that the number of collaborations increased between 0.3 and 1.1 times and stimulated more novel projects of higher quality. However, those opportunities do not equally benefit individuals. Specifically, female scientists do not respond to lower travel cost incentives because women ‘may have more constrained travel schedules’ than men ([Catalini, Fons-Rosen, and Gaulé 2020](#)).

[Jones \(2021\)](#) discusses more broadly the problem of having access to teams and claims that biases might limit the selection of specific categories of persons in teams, mainly when the team members’ screening happens among people who do not know each other personally. One potentially important bias is the one against women. In the growing debate about gender equality in science ([Larivière et al. 2013](#); [Holman, Stuart-Fox, and Hauser 2018](#)), understanding gender bias in team formation becomes crucial because the existence of such a gender bias would be detrimental to women’s careers and the entire society when potential talents are limited or excluded. Extant studies have analyzed the role played by gender in team formation, looking at the coauthorship in published papers. For instance, [Boschini and Sjögren \(2007\)](#) analyze coauthorships in papers published in three top economics journals from 1991 to 2002 and find that team formation is not gender-neutral. They show that even when women are well represented in the field, men tend not to team up with them. ‘Women are more than twice as likely as men to have a female coauthor’ (338). Women are also more likely to publish solo-author papers than their male colleagues, and ‘as the fraction of female researchers [in the field] increases, women increasingly tend to write with other women’ (339). As [Boschini and Sjögren \(2007\)](#), we choose academia as an empirical setting since, in this context, team formation is voluntary. Scientists choose the colleagues they want to work with based on individual preferences and expectations. Unlike [Boschini and Sjögren \(2007\)](#), we consider scientists who team up to apply for a grant, regardless of the result of the grant competition. By doing that, we overcome the limitation of analyzing only successful teams where the bias observed might result from a selection process discriminating against certain types of teams. We cover all the disciplines and implement a quasi-natural experiment comparing the proportion of women in the observed teams with those in a control group of potential candidates. As ([Card et al. 2023](#)) pointed out, fewer women observed in teams might result from a scarcity of women among qualified candidates or from a bias in selecting between equally qualified women and men. The quasi-natural experiment design described in [Section 3](#) aims to disentangle those two cases to identify a bias in the selection of female teammates.

3. Empirical strategy

Using data on the observed teams, we estimate the probability of observing a female (F) PI conditional on having joined a team ($join$), $P(F|join)$. We do so by calculating the probability of observing a female PI in our study sample, $P(F|join)$. (We adopt a frequentist definition of sample probability calculating the ratio between the number of female researchers in our observed teams and the total number of researchers in our observed teams. Equivalently, we can also state that the

probability of observing a female researcher in our sample is calculated as the arithmetic average, as indicated by the upper bar in $\overline{P(F|join)}$, of a dummy variable that takes the value of 1 if the researcher is a female researcher, 0 otherwise.) However, the pool of potential candidates from which the observed ones are selected can have a different gender composition depending on the discipline and abilities required. Having no female scientists in the team might not necessarily reflect a gender selection bias, but a lack of female scientists in the discipline or a lack of women having the competencies needed for the research project. For instance, a low share of women in historically male-dominated disciplines such as physics and mathematics ([Stephan 1996](#); [Ginther and Kahn 2004](#)) might not denote a gender bias but simply a scarcity of women in the eligible pool of candidates. In other fields, like psychology, where women are highly represented, a low share of women in teams might be associated with a selection bias ([Card et al. 2023](#)).

To account for the difference in the share of potentially qualified women and correctly estimate the existence of a gender bias (if any) in teammate selection, we compare the probability of having a female researcher in the observed teams, $P(F|join)$, with the probability of having a female researcher in a matched control group of individuals similar to those selected according to their observable characteristics. (The characteristics of the observed PIs correspond to the revealed preferences of both PLs and PI for starting a scientific collaboration. For this reason, our control group is made by scientists similar to the observed PIs. For instance, by also including as potential control individuals in the same department as the observed PIs, we take into account PLs and PIs’ preferences for the geographical location. In other words, if a PL teamed up with a PI affiliated with a specific department, we assume that the observed PI’s departmental colleagues (our potential controls) are willing to sustain the commuting costs associated with starting a collaboration with the PL.) Specifically, we estimate the unconditional probability of observing a female researcher $P(F)$ in the pool of potentially qualified PIs by calculating the probability of observing a female PI in our control group, $\overline{P(F)}$. Constructing the control group is needed because comparing the probability of having a female researcher in the observed teams, i.e. $P(F|join)$, with the probability of observing a female researcher in the entire scientific community would have introduced a bias in our estimations. Indeed, researchers in the scientific community might have different characteristics (other than gender) than those selected by the PLs. We use a two-tailed statistical test to assess if the probabilities of having a female scientist among the PIs in the observed teams and the pool of potential PIs are statistically equivalent (H_0) or statistically different (H_1). [Equation \(1\)](#) formalizes the null and alternative hypotheses:

$$\begin{aligned} H_0 : P(F|join) &= P(F) \\ H_1 : P(F|join) &\neq P(F) \end{aligned} \quad (1)$$

To detail further our results, we calculate the ratio $\overline{P(F|join)}/\overline{P(F)}$ to assess if female researchers are more (or less) frequent among individuals in the observed teams than in the control group. (An easily interpretable equivalent of the calculated ratio results from applying the Bayes rule. Indeed, we can interpret r as the ratio between the probability of joining a team conditional on the PI’s gender and the unconditional probability of joining a team, i.e. $P(join|F)/P(join)$.) In other

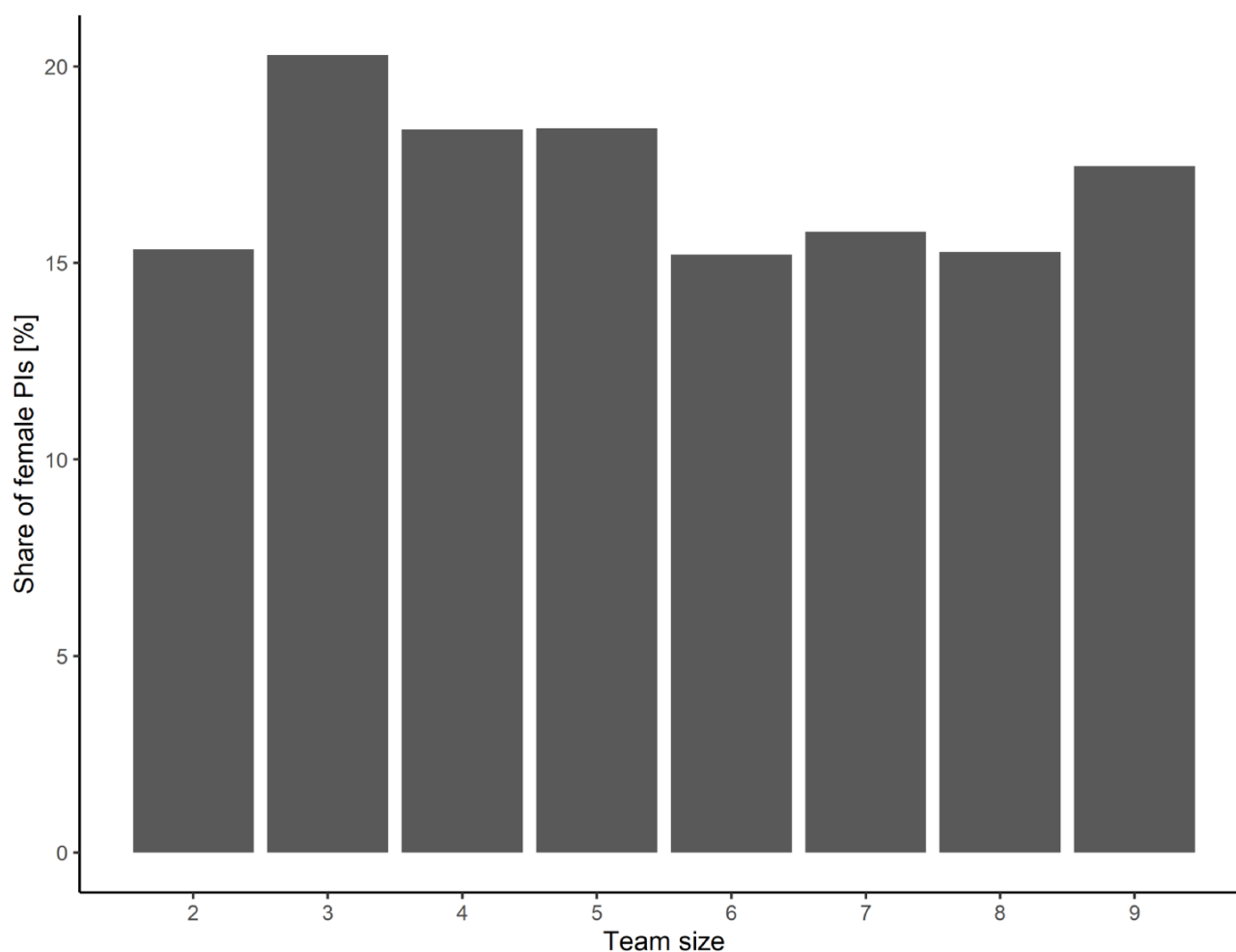


Figure 1. The average share of female scientists by the team size.

words, if the ratio $\overline{P(F|join)}/\overline{P(F)}$ is lower than 1, a gender bias against selecting female teammate scientists exists, while if the ratio is higher than 1, a gender bias in favor of selecting female scientists exists.

Since assessing the existence of a gender bias in team formation crucially depends on the reliability of the control group, defining the pool of qualified potential PIs is the main challenge of our empirical analysis. [Section 4](#) describes our data and details how we identify the control group of potential PIs.

4. Data

The ESF is a nongovernmental, nonprofit foundation that promotes high-quality research projects in Europe. Among the ESF's funding opportunities, the European Collaborative Research initiative (EUROCORES) was launched to promote an 'independent, bottom-up approach to collaborative research in Europe that allowed new scientific ideas to be developed' (ESF 2015; Bianchini *et al.* 2022). One of the EUROCORES objectives was to sponsor collaborations that would not have taken place without such funds. A team led by a PL and a minimum of two PIs originating from two different countries was eligible to apply with a project to the EUROCORES' open call on a broad range of disciplines, including life and medical sciences, physics and engineering, and humanities.

4.1 Observed teams

In our analysis, we consider 329 teams applying with a project to the ESF between 2006 and 2010. The team size ranges from two to nine PIs, with an average team of 3.83 PIs. Most teams count two or three PIs, 26.7 and 28.0 per cent of the teams, respectively. Only 17.6 per cent of the teams have more than five PIs, and 2.13 per cent of the teams count nine members. Germany is the country that benefits the most from the ESF's funds (see [Appendix Table A.1](#) for the distribution of PIs and PLs by country). The average share of female scientists does not vary substantially by the team size ([Fig. 1](#)) or application year ([Fig. 2](#)). On average, each team has 17.5 per cent of female PIs.

For each application, we obtained from the ESF information on the name of the applicants, gender, and affiliations. We complemented that information with publication records retrieved from Scopus Elsevier's database. [Table 1](#) reports the main descriptive statistics for our sample of 1,260 PIs. On average, PIs are senior scholars with a solid academic curriculum. At the application time, the average PI has 15 years of experience from the first publication, reports a stock of forty publications, and leverages an extensive network of coauthors (87.19). When considering the distribution of the PI–PL gender pairs, female PIs tend to be led more often by female PLs (21 per cent) than by male PIs (12 per cent).

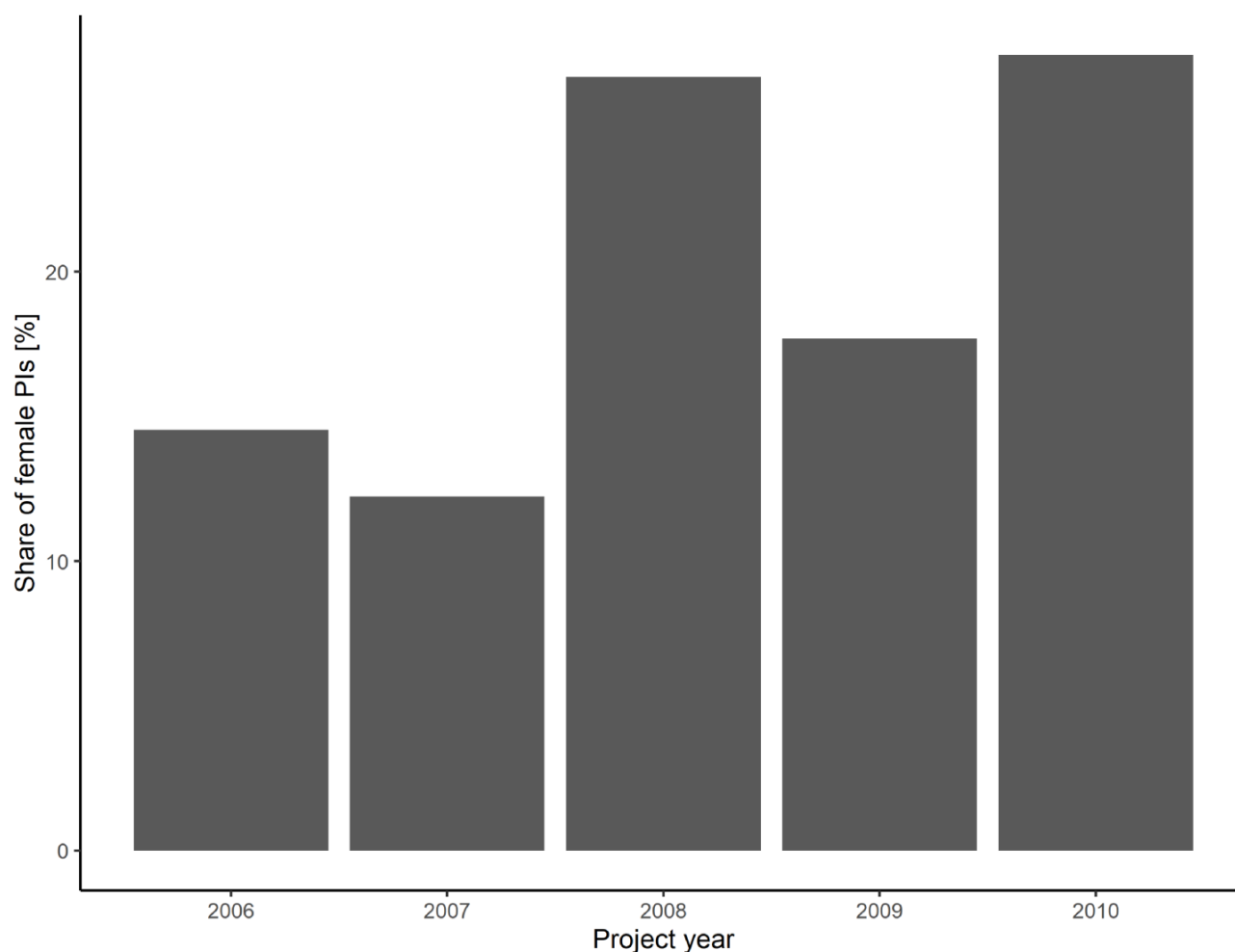


Figure 2. The average share of female scientists by the application year.

Table 1. Profiles of the PIs observed in our sample at the application time, by gender.

	Observed PIs (1,260)			
	Female PIs (221)		Male PIs (1,039)	
	Average	SD	Average	SD
Number of publications	40.48	41.19	74.98	96.37
Career length	15.70	8.33	19.23	9.67
Number of coauthors	87.19	95.66	121.86	152.41
Female PL (dummy variable)	0.21	0.41	0.12	0.32

4.2 Defining the control group

To define our control group of qualified PIs, we follow a three-step procedure. First, we construct a large pool of researchers with the potential to be selected as PIs by the PLs (Step 1). Then, among those potential PIs, we extract those with competencies aligned with the project application team (Step 2). Finally, we match observed PIs with potential PIs identified

in Steps 1 and 2, relying on observable characteristics such as publication profile, career length, and scientific network (Step 3). By doing that, we identify a ‘twin’ for each observed PI, i.e. a scientist with similar characteristics (excluding gender) who could have joined the team but did not.

4.2.1 Identifying the pool of potential controls

As Step 1, we construct the group of potential PIs assuming that PLs choose their teammate PIs within three different pools of researchers: (1) those researchers with whom they have collaborated in the past, (2) colleagues of whom they know the work, and (3) researchers working in departments conducting research on subjects functional to the project they intend to propose to the ESF.

To reconstruct those three pools, we collect (1) the list of coauthors with whom the PLs have worked before applying for an ESF call (*PLs coauthors*), (2) the list of authors cited by the PLs in the works published before applying for the ESF call (*PLs cited authors*), and (3) the list of departmental colleagues of the PIs selected who are active before the application date of the ESF call (*Colleagues of the observed PIs*).

4.2.1.1 PLs coauthors. Looking at the coauthors’ list with whom the 329 PLs of our sample collaborate, we find, on average, 62.86 coauthors for each PL.

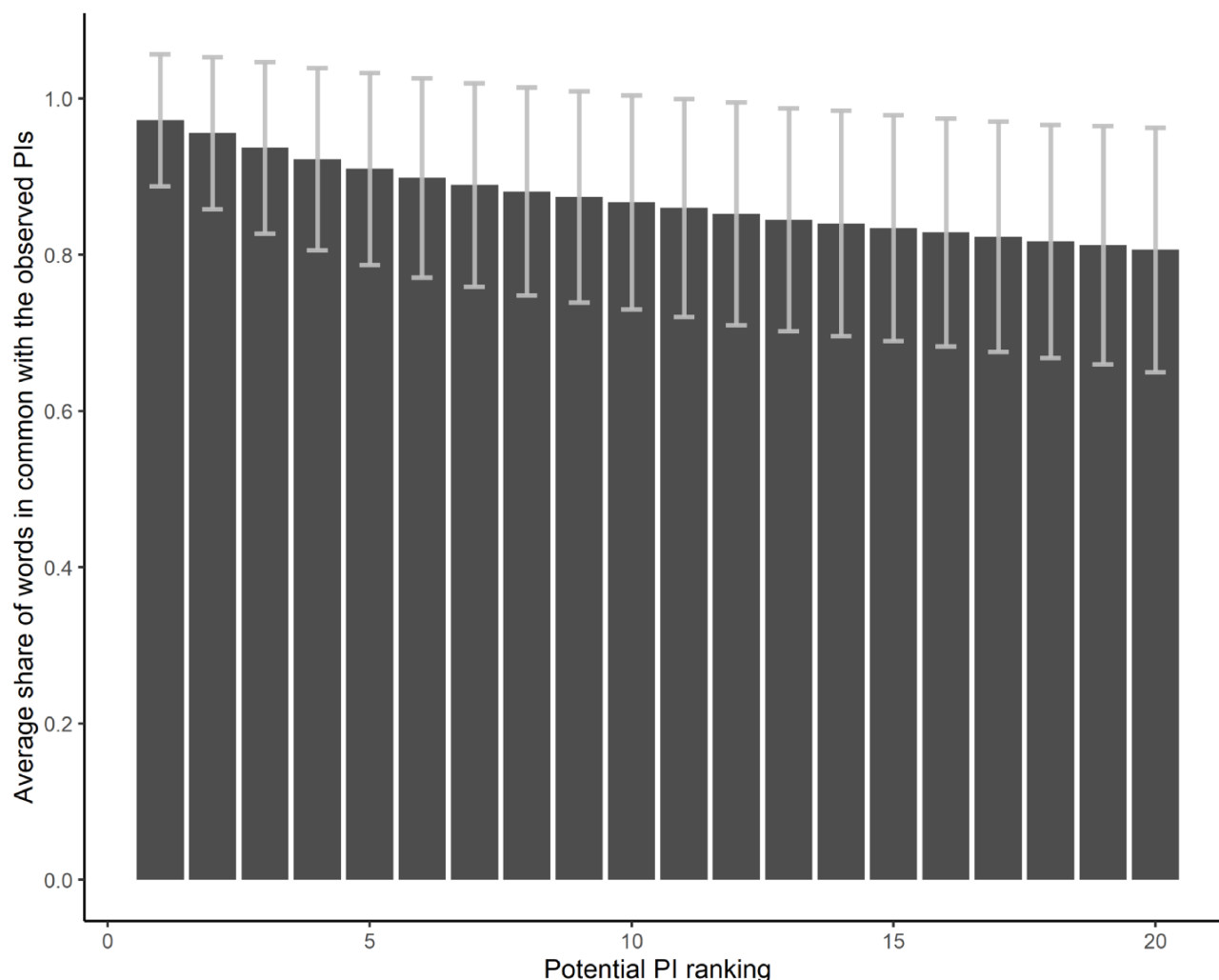


Figure 3. The average share of words in common with the observed PIs according to the ranking of the potential PIs, from the one with the highest to the lowest share of common words. *Note:* Error bars represent 1 SD from the average.

4.2.1.2 PL's cited authors. In identifying the authors cited in PLs' articles, we limit our search to papers with less than twenty references (in counting the references, we consider only research papers with a DOI; we also exclude books and book chapters) listed to exclude literature reviews. We also excluded the authors of papers older than 10 years at the project application date to ensure that the authors considered are likely to be active researchers. On average, the PLs in our sample have 81.12 distinct authors in their reference lists.

4.2.1.3 Colleagues of the observed PIs. Since PLs are supposed to target researchers in departments conducting research in the field of interest for their projects, we assume that the departments of the selected PIs are potential sources of talent functional to the project. Then, we identify as potentially talented PIs those scientists who are departmental colleagues of the selected PIs, i.e. those working at the same university and discipline as the selected PIs in the year of the project application. Using the Scopus database, we create the list of departmental colleagues, extracting authors'

names with the same affiliation and discipline as the observed PIs. We defined the discipline by identifying the observed PIs' Scopus subject area in which they published the most. On average, we identify for each PL in our sample 80.62 distinct authors who are departmental colleagues of the selected PIs.

After pooling together these three lists of scientists, i.e. coauthors, cited authors, and departmental colleagues, we identified 69,710 potential PIs for the 329 teams, with, on average, 211.89 potential PIs for each team. Among the PIs part of the control group, 32 per cent are previous coauthors of the PL, 36 per cent are scientists listed in the references of the PL publications, and 37 per cent are departmental colleagues of the selected PIs. As the last step, we assigned to each potential PI the gender using the information gathered from the Gender Name Dictionary of the World Intellectual Property Organization (Martínez, Raffo, and Saito 2016). In the pool of the 69,710 potential PIs, 27 per cent are women.

4.2.2 Matching the controls

In the pool of potential PIs, each PL selects her teammates by looking closely at their scientific profiles. We assume that the observed PIs have the ideal characteristics the PL is looking for. To measure gender bias in the team selection, we extract from the large pool of potential PIs those scientists whose profiles match the ones of the selected PIs along a set of relevant characteristics excluding gender. To do that, we implement the remaining two steps of our procedure. In Step 2, we identify the twenty potential PIs with the most similar scientific competencies to the observed PIs. By assuming that scientific competencies are disclosed in publications, we select all the articles published by the observed PIs before the project application date and extract the list of words (we excluded stop words) appearing in titles and abstracts. Similarly, we isolate the words appearing in the title and abstract of the potential PIs' articles. Once each observed and potential PI is associated with a bag of words, we ranked the potential PIs according to the number of words shared with the observed PIs for each project. Figure 3 shows on the horizontal axis the ranking of potential PIs according to their share of words and, on the vertical axis, the average share of common words for each ranking position across teams. For instance, the first bar on the left of the graph (*Potential PI ranking* = 1) represents the average share of common words between the observed PIs in our 329 teams and the potential PI with the highest share of common words. From Fig. 3, it emerges that if the ranking position is lower than 20, the average value of the share of common words remains higher than 0.8, meaning that the potential PI shares with the observed PIs more than 80 per cent of the words used in her papers' titles and abstracts. To guarantee that the observed and potential PIs are scientists having competencies in the same subjects, we select for each project the twenty PIs with the highest share of common words with the observed PIs. By doing so, we restrict our control group from 211.89 potential PIs per project to 20 potential PIs with similar competencies to the selected ones. Looking at the origin of the researchers in our control group, 39 per cent of the researchers have coauthored with the PL (*PL's coauthors pool*), 32 per cent have been cited by the PL (*PL's cited authors pool*), and 36 per cent are departmental colleagues of the observed PIs (*Colleagues of the observed PIs pool*). (The percentages do not sum up to 100 percent since one potential PI might be selected according to more than one criterion.)

In Step 3, among the twenty potential PIs selected by the project, we conducted a matching exercise with the observed PIs. Specifically, for each observed PI, we identify the nearest potential PI in terms of productivity, career length, and number of coauthors until the year of the project application. To do so, we construct three categorical variables. The first variable *High productivity* is a dummy variable that equals 1 if the PI has a number of publications higher than the median value in our sample, i.e. nineteen publications at the time of the project application. The second dummy variable, *Long career*, considers the career's length. We define *Long career* as a dummy variable that equals 1 if the time elapsed from the date of the first PI's publication to the year of the application is greater than 13 years, where 13 is the median career length in our sample. Finally, we define the dummy variable *Large network* as a variable that equals 1 if the PI has a large coauthorship network, i.e. more than the median number of thirty-nine coauthors in our sample.

Table 2. Propensity score matching regression: logit estimates and marginal effects reported.

	(1) Observed PI
High productivity (dummy variable)	0.15*** (0.014)
Long career (dummy variable)	0.028** (0.011)
Large network (dummy variable)	0.034*** (0.012)
Missing career information (dummy variable)	0.13*** (0.041)
Observations	7,840 (1,260 + 6,580)
Pseudo- R^2	0.0728

Notes: We do not have the information concerning the PIs' careers for 1 per cent of our sample. In these cases, we set the dummy variable *Long career* equal to 0, and we calculate a dummy variable *Missing career information* that equals 1 when the career information is missing, 0 otherwise. Standard errors are reported in parentheses.

* $P < 0.1$.

** $P < 0.05$.

*** $P < 0.01$.

Table 3. Comparison between observed and potential PIs' characteristics.

	Observed PIs	Potential PIs	H0: $\mu_{\text{Observed}} = \mu_{\text{Potential}}$ P-value
High productivity (dummy variable)	0.78	0.76	0.14
Long career (dummy variable)	0.72	0.70	0.29
Large network (dummy variable)	0.72	0.71	0.62

Table 2 reports the matching equation estimated with a logistic regression where the explained variable is the dummy variable *Observed PI*, which equals 1 if the PI belongs to the list of the 1,260 observed PIs and 0 if the PI belongs to the list of the 6,580 potential PIs (6,580 potential PIs correspond to 20 potential PIs for each of the 329 project applications, i.e. $20 \times 329 = 6,580$). The estimates show that high-productive PIs with long careers and large networks are more likely to become part of the observed teams.

As a result of Step 3, we match 1,260 observed PIs with 1,260 potential PIs. Table 3 compares the average values of the three matching variables for the observed and potential PIs, showing the quality of the matching exercise. Specifically, 78 per cent of the observed PIs have high productivity compared to 76 per cent of the potential PIs. Seventy-two percent of the observed PIs have a long career compared to 70 per cent of the potential PIs. Finally, 72 per cent of the observed PIs have a large network compared to 71 per cent of the potential PIs. All the differences between the two groups of PIs, i.e. observed and potential, are not statistically significant.

Table 4 summarizes the average number of potential PIs for each step of the three-step procedure implemented to construct our control group and the share of potential PIs drawn from each pool. To validate our potential PIs' selection procedure, in Appendix 2, we assess the share of observed PIs that

Table 4. The average number of controls per project application along the three-step procedure implemented to identify the control group of potential PIs and the share of researchers drawn from each pool.

	Pool 1: PL's coauthors (%)	Pool 2: PL's cited authors (%)	Pool 3: colleagues of the observed PIs (%)	Average number of potential PIs per project
Step 1: controls in Pools 1, 2, and 3	32	36	37	211.89
Step 2: controls with the competencies needed for the project	39	32	36	20
Step 3: controls after propensity score matching	39	34	35	3.83 (one for each observed PI)

Notes: The average number of PI controls per project reported for Step 3 (3.83) equals the average number of observed PIs. These values are expected because we retrieved one potential PI for each observed PI.

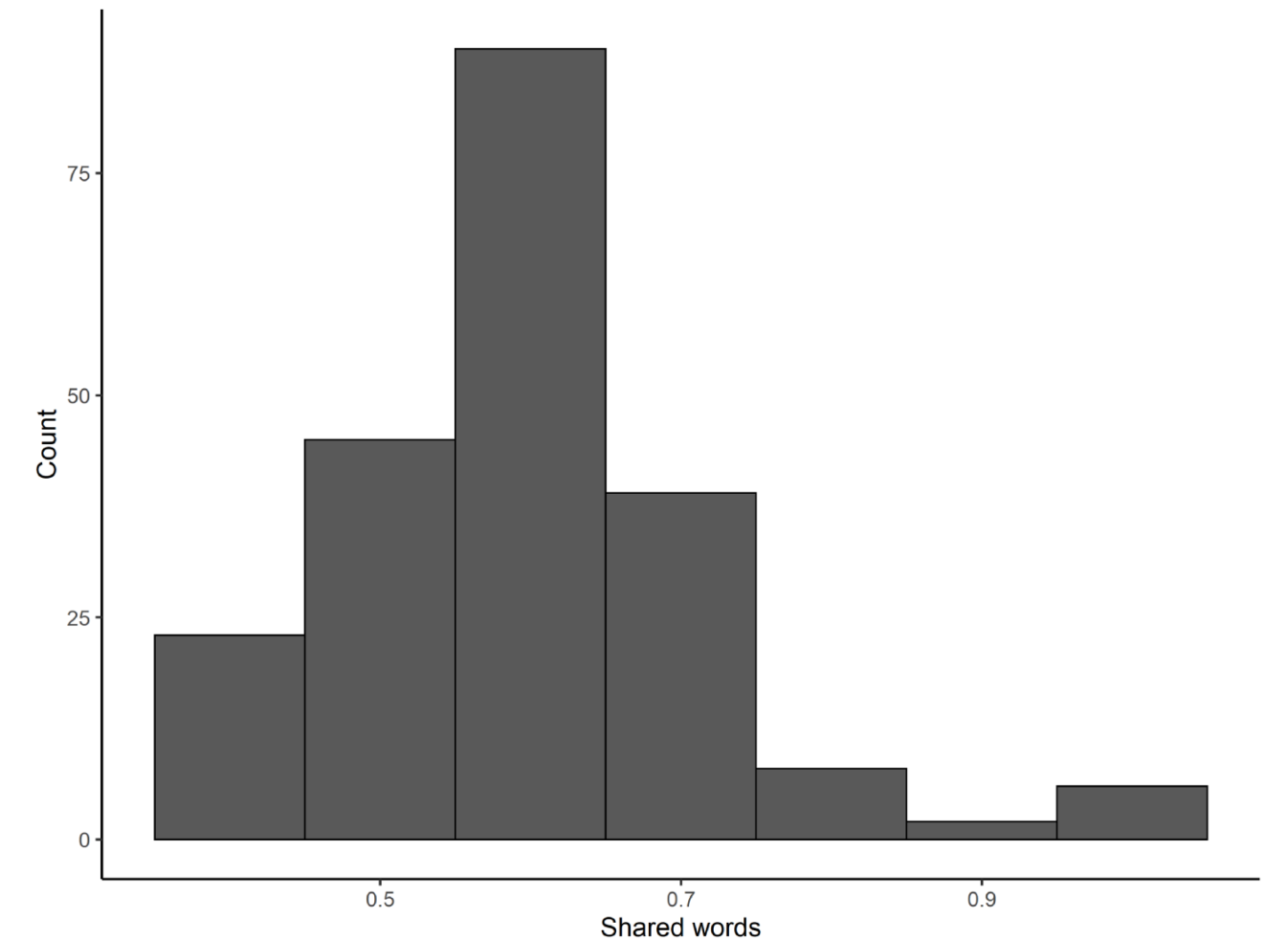


Figure 4. Distribution of the share of words in common between the vector of the competencies needed for project X and the vectors representing the articles where we found the potential PIs.

belongs to each of the three pools from which we assume the PL draws for inviting teammates. We find that about half of the observed PIs are drawn from these pools, showing that they are a reliable source of teammates for the PL.

4.2.3 An illustrative example

To illustrate our three-step procedure, we present in this section an anonymized example. Project application X was crafted in 2009 and listed one male PL, PL1, affiliated with the

University of Turku in Finland. The same application included two male PIs, PI1 and PI2, affiliated with the University of Jyväskylä in Finland and the Konrad Lorenz Institute in Austria, respectively. The project application was in the field of ethology, the study of animal behavior. PL1 had eighty-eight publications before 2009, listing eighty-one distinct coauthors and 184 referenced authors in their bibliographies. (These figures are the results of applying the constraints defined in Section 4.2.1. For instance, in searching for referenced

authors, we consider only papers with less than twenty references listed to exclude literature reviews. We also excluded the authors of papers older than 10 years at the project application date to ensure that the authors considered are likely to be active researchers.) Moreover, PI1 and PI2 have forty-one colleagues publishing in the same field at the Konrad Lorenz Institute (thirteen colleagues) and the University of Jyväskylä in 2009 (twenty-eight colleagues). Overall, we identified 283 potential PIs for this project application X. The number of potential PIs is not the algebraic sum of the eighty-one coauthors, 184 referenced authors, and forty-one PIs' departmental colleagues because roles might overlap. Indeed, a scientist might be a coauthor, referenced author, or PIs' departmental colleague simultaneously. Specifically, 28.6 per cent of the 283 potential PIs are PL's coauthors, 65.0 per cent are referenced authors, and 14.5 per cent are PIs' departmental colleagues.

Once we identified the 283 potential PIs, we retrieved the articles' titles and abstracts in which they were listed as authors and created the 283 corresponding vectors of words gathered from those titles and abstracts. Then, we calculated a similar vector, including all the words in the articles' titles and abstracts published by PI1 and PI2 before the application date. This latter vector represents the competencies shown by the PIs in contributing to the project. Figure 4 illustrates the distribution of the share of words in common between the vector representing the competencies required for the project and each of the 283 vectors of words representing the content of the articles where the potential PIs were listed. Finally, we selected the potential PIs with the twenty highest shares, retrieving twelve female and eight male researchers. Among those researchers, we selected the most similar to PI1 according to three observable characteristics: productivity, career, and network size. We did the same for PI2. For PI1, we found a female researcher with similar characteristics (control C1), while for PI2, we found a male researcher (control C2). In this example, a male researcher PI1 has been invited by PL1 to join the application team, but potentially a female researcher with the same characteristics (C1) could have been chosen. We can say that PL1 might have chosen PI1 as a teammate instead of C1 due to gender bias in selection.

5. Results

To identify gender bias in team selection, we compare the probability of having a female scientist in the teams of observed PIs, $P(F|join)$, with the probability of having a female scientist in the control group of potential PIs identified with the three-step procedure illustrated earlier, $P(F)$. Specifically, as illustrated in Section 3, we first test if the two probabilities are statistically equivalent (H_0) or statistically different (H_1). Then, we compute the ratio $\overline{P(F|join)/P(F)}$. A ratio $\overline{P(F|join)/P(F)}$ equal to 1 means no gender bias in team selection since the probability of having a female scientist in the observed team equals the probability of having a female scientist in the control group.

Results reported in Table 5 show gender bias against female PIs when PLs form their teams. When we test the null hypothesis that $P(F|join) = P(F)$, we obtain a P -value of 0.001, below the standard significance level of 0.01. This result allows us to reject the null hypothesis that the probability of having a female PI in the observed teams equals the probability of

Table 5. Comparing $P(F|join)$ and $P(F)$ to assess gender bias.

	Probabilities, P -values, and ratios
$\overline{P(F join)}$	17.5% (1,260 observed PIs)
$\overline{P(F)}$	22.6% (1,260 potential PIs)
$H_0 : P(F join) = P(F);$ $H_1 : P(F join) \neq P(F)$	0.001 (P -value)
$\overline{P(F join)/P(F)}$	0.77

having a female PI in the control group of potential PIs. In other words, the choice of PIs is not gender-neutral. Indeed, the ratio $\overline{P(F|join)/P(F)}$ equals 0.77 being the proportion of female PIs in the observed teams 17.5 per cent, $\overline{P(F|join)}$, and the one in the sample of potential PIs 22.6 per cent, $\overline{P(F)}$.

5.1 Exploring teams' heterogeneity

Table 5 compares the probability of observing female scientists in the samples of observed and potential PIs without distinguishing the typology of teams to which those PIs belong. A possible concern is that gender bias might differ in teams with different characteristics. In this section, we explore gender bias team heterogeneity along seven dimensions such as (1) PL's gender, (2) PL's publication productivity, (3) PL's seniority, (4) PL country's masculinity score, (5) project discipline, (6) result of the funding decision, and (7) team's cohort:

- (1) PL's gender: when we split our sample by the gender of the PL, we find that the gender bias against selecting female PIs is driven by male PLs (Table 6). For the 1,089 PIs led by a male PL, when we test the hypothesis that $P(F|join) = P(F)$, we obtain a P -value close to 0, rejecting the hypothesis for the standard significance level of 0.01. The ratio $\overline{P(F|join)/P(F)}$ equals 0.69 depicting a gender bias against female PIs. On the contrary, for the 171 PIs in teams led by a female PL, when we test the hypothesis that $P(F|join) = P(F)$, we obtain a P -value of 0.129, which does not allow us to reject the null hypothesis that $P(F|join) = P(F)$ for the standard significance level of 0.1. Therefore, based on our empirical evidence, we cannot conclude that teams led by female PLs are gender biased. Nonetheless, the high P -value might be due to the non-negligible reduction in the observations used in the statistical test when considering only teams led by female PLs. Indeed, the test is conducted by comparing a sample of 171 observed PIs and 171 potential PIs, a sample size of almost one-tenth of the original sample of 1,260 observed PIs and 1,260 potential PIs.
- (2) PL's publication productivity: PLs with different productivity levels might behave differently when inviting PIs to join the team. We distinguish between high- and low-productive PLs according to their publication records before the application date of the project. We define a PL as high productive if she has a number of publications higher than the median number of publications of the PLs in our study sample, while as low productive otherwise. Table 7 shows that, for both high- and low-productive PLs, we can reject the null hypothesis that $P(F|join) = P(F)$ according to the 0.1 threshold level. However, when we look at the ratio

Table 6. Comparing $P(F|join)$ and $P(F)$ to assess gender bias distinguishing teams led by male and female PLs.

Probabilities, P -values, and ratios	
Conditional on having a female PL	
$\overline{P(F join)}$	27.5% (171 observed PIs)
$\overline{P(F)}$	20.5% (171 potential PIs)
$H_0 : P(F join) = P(F); H_1 : P(F join) \neq P(F)$	0.129 (P -value)
$\overline{P(F join)}/\overline{P(F)}$	1.34
Conditional on having a male PL	
$\overline{P(F join)}$	15.8% (1,089 observed PIs)
$\overline{P(F)}$	22.9% (1,089 potential PIs)
$H_0 : P(F join) = P(F); H_1 : P(F join) \neq P(F)$	0.000 (P -value)
$\overline{P(F join)}/\overline{P(F)}$	0.69

Table 7. Comparing $P(F|join)$ and $P(F)$ to assess gender bias distinguishing teams led by high- or low-productive PLs.

Probabilities, P -values, and ratios	
Conditional on having a high-productive PL	
$\overline{P(F join)}$	18.6% (630 observed PIs)
$\overline{P(F)}$	23.0% (630 potential PIs)
$H_0 : P(F join) = P(F); H_1 : P(F join) \neq P(F)$	0.052 (P -value)
$\overline{P(F join)}/\overline{P(F)}$	0.807
Conditional on having a low-productive PL	
$\overline{P(F join)}$	16.5% (630 observed PIs)
$\overline{P(F)}$	22.2% (630 potential PIs)
$H_0 : P(F join) = P(F); H_1 : P(F join) \neq P(F)$	0.0102 (P -value)
$\overline{P(F join)}/\overline{P(F)}$	0.743

$\overline{P(F|join)}/\overline{P(F)}$, we observe that the gender bias is more pronounced in those teams led by low-productive PLs (0.743 versus 0.807).

- (3) PL's seniority: junior and senior PLs might behave differently when choosing their PIs. We distinguish between senior and junior PLs according to the number of years that elapsed from their first publication. We define as seniors those PLs with a number of years of experience greater than the median in our study sample, while juniors are the remaining. Table 8 shows that only for senior PLs, we can reject the hypothesis that $P(F|join) = P(F)$ according to the 0.01 threshold level. For teams led by juniors, when we test the hypothesis that $P(F|join) = P(F)$, we obtain a P -value of 0.1784, which does not allow us to reject the null hypothesis for the standard significance level of 0.1. Therefore, for teams led by senior PLs, the ratio $\overline{P(F|join)}/\overline{P(F)}$ equals 0.721, denoting a gender bias against women, while, for junior PLs, we do not observe any statistically significant gender bias.
- (4) PL country's masculinity score: the PL's cultural background might affect her behavior when asking PIs to join the team. To proxy for the cultural background of the PL, we follow Hofstede and Hofstede (2001) and look at the masculinity score associated with the country where the PL is based. We distinguish between

Table 8. Comparing $P(F|join)$ and $P(F)$ to assess gender bias distinguishing junior and senior PLs.

Probabilities, P -values, and ratios	
Senior PL	
$\overline{P(F join)}$	17.22% (691 observed PIs)
$\overline{P(F)}$	23.88% (691 potential PIs)
$H_0 : P(F join) = P(F); H_1 : P(F join) \neq P(F)$	0.0022 (P -value)
$\overline{P(F join)}/\overline{P(F)}$	0.721
Junior PL	
$\overline{P(F join)}$	17.92% (569 observed PIs)
$\overline{P(F)}$	21.09% (569 potential PIs)
$H_0 : P(F join) = P(F); H_1 : P(F join) \neq P(F)$	0.1784 (P -value)
$\overline{P(F join)}/\overline{P(F)}$	0.850

Table 9. Comparing $P(F|join)$ and $P(F)$ to assess gender bias distinguishing teams led by PLs based in high and low-masculine countries.

Probabilities, P -values, and ratios	
Conditional on having a PL based in a high masculine country	
$\overline{P(F join)}$	16.3% (771 observed PIs)
$\overline{P(F)}$	21.3% (771 potential PIs)
$H_0 : P(F join) = P(F); H_1 : P(F join) \neq P(F)$	0.0133 (P -value)
$\overline{P(F join)}/\overline{P(F)}$	0.768
Conditional on having a PL based in a low-masculine country	
$\overline{P(F join)}$	19.4% (489 observed PIs)
$\overline{P(F)}$	24.7% (489 potential PIs)
$H_0 : P(F join) = P(F); H_1 : P(F join) \neq P(F)$	0.0451 (P -value)
$\overline{P(F join)}/\overline{P(F)}$	0.785

Notes: We calculate the median masculinity score value across countries, regardless of the number of projects by country in our study sample. For this reason, the number of PIs in highly masculine countries (771 observed + 771 potential PIs) differs from that in low-masculine countries (489 observed + 489 potential PIs).

high masculine countries where gender roles are distinct and low-masculine countries where social gender roles overlap. We define high masculine countries as the ones for which the masculinity score value is above the median value in our study sample, while we define low-masculine countries otherwise. Table 9 shows that in both cases, there is gender bias in team formation. However, teams led by a PL based in a high masculine country show a slightly higher gender bias against women than PLs based in low-masculine countries, i.e. the ratio $\overline{P(F|join)}/\overline{P(F)}$ equal to 0.768, a lower value than in low-masculine countries.

- (5) Project discipline: when looking at the results presented in Table 5, a possible concern is that gender bias might differ across disciplines. To address this concern, we split the applications by discipline identifying applications in *physics and engineering*, *life and medical sciences*, and *humanities*. Table 10 shows that gender bias against female scientists affects *life and medical sciences* and *humanities*, while it is not present in *physics and engineering*. Specifically, in *life and medical sciences*, the ratio $\overline{P(F|join)}/\overline{P(F)}$ equals 0.807 with a P -value of 0.0375 for the test of the hypothesis $P(F|join) = P(F)$,

Table 10. Comparing $P(F|join)$ and $P(F)$ to assess gender bias in different disciplines.

Probabilities, P -values, and ratios	
Physics and engineering	
$\overline{P(F join)}$	13.9% (332 observed PIs)
$\overline{P(F)}$	14.2% (332 potential PIs)
$H_0 : P(F join) = P(F); H_1 : P(F join) \neq P(F)$	0.911 (P -value)
$\overline{P(F join)}/\overline{P(F)}$	0.979
Life and medical sciences	
$\overline{P(F join)}$	18.9% (709 observed PIs)
$\overline{P(F)}$	23.4% (709 potential PIs)
$H_0 : P(F join) = P(F); H_1 : P(F join) \neq P(F)$	0.0375 (P -value)
$\overline{P(F join)}/\overline{P(F)}$	0.807
Humanities	
$\overline{P(F join)}$	19.8% (232 observed PIs)
$\overline{P(F)}$	31.5% (232 potential PIs)
$H_0 : P(F join) = P(F); H_1 : P(F join) \neq P(F)$	0.004 (P -value)
$\overline{P(F join)}/\overline{P(F)}$	0.630

Table 11. Comparing $P(F|join)$ and $P(F)$ to assess gender bias distinguishing awarded and non-awarded projects.

Probabilities, P -values, and ratios	
Conditional on the application being awarded	
$\overline{P(F join)}$	19.11% (492 observed PIs)
$\overline{P(F)}$	24.6% (492 potential PIs)
$H_0 : P(F join) = P(F); H_1 : P(F join) \neq P(F)$	0.04 (P -value)
$\overline{P(F join)}/\overline{P(F)}$	0.78
Conditional on the application not being awarded	
$\overline{P(F join)}$	16.5% (768 observed PIs)
$\overline{P(F)}$	21.4% (768 potential PIs)
$H_0 : P(F join) = P(F); H_1 : P(F join) \neq P(F)$	0.02 (P -value)
$\overline{P(F join)}/\overline{P(F)}$	0.77

Table 12. Comparing $P(F|join)$ and $P(F)$ to assess gender bias distinguishing project applications in 2006–7 and 2009–10.

Probabilities, P -values, and ratios	
Conditional on the application being in 2006–7	
$\overline{P(F join)}$	13.40% (679 observed PIs)
$\overline{P(F)}$	20.91% (679 potential PIs)
$H_0 : P(F join) = P(F); H_1 : P(F join) \neq P(F)$	0.00024 (P -value)
$\overline{P(F join)}/\overline{P(F)}$	0.64
Conditional on the application being in 2009–10	
$\overline{P(F join)}$	21.29% (465 observed PIs)
$\overline{P(F)}$	23.87% (465 potential PIs)
$H_0 : P(F join) = P(F); H_1 : P(F join) \neq P(F)$	0.347 (P -value)
$\overline{P(F join)}/\overline{P(F)}$	0.89

while in *humanities*, the ratio $\overline{P(F|join)}/\overline{P(F)}$ equals 0.630 with a P -value of 0.004 for the test of the hypothesis $P(F|join) = P(F)$. A possible interpretation of these findings is that the few women in physics and

engineering attain the appreciation of their colleagues for having succeeded in a male-dominated field. Making a career in physics and engineering is perceived as a signal of high quality and commitment to work or to be aligned with a masculine role model (Stephan 2012).

- (6) Result of the funding decision: the results in Table 5 might raise the possible concern that we are grouping projects of different quality when considering together awarded and non-awarded applications. It might be that awarded projects have a different team composition than non-awarded projects. To address this concern, we split the applications by awarded status. Table 11 shows that when we test the hypothesis that $P(F|join) = P(F)$ for the two samples, we obtain both P -values below the standard significance level of 0.05, allowing us to reject the null hypothesis of equal probability of having female and male PIs. The gender bias against female PIs is similar in awarded and non-awarded projects. The ratio $\overline{P(F|join)}/\overline{P(F)}$ equals 0.78 for the awarded projects and 0.77 for non-awarded projects.
- (7) Team's cohort: recent studies have shown that the gender bias against female scientists in team formation has varied over time (Card et al. 2023). To analyze the time dynamics, in Table 12, we test for gender bias in team formation in two time windows of 2 years each, i.e. 2006–7 and 2009–10, covered by our study sample. Table 12 shows that in recent years, the gender bias against women selection has substantially reduced.

6. Conclusions

Teamwork has become the dominant model for producing science (Wuchty, Jones, and Uzzi 2007). Scientists team up to create new knowledge by joining their competencies and efforts. Moreover, teamwork stimulates learning and creativity, benefiting individual teammates (Ayoubi, Pezzoni, and Visentin 2017). Funding agencies have also recognized the advantage of teamwork and included dedicated funding schemes reserved for team applications in their funding portfolios. Several studies have analyzed the impact of team composition on performance once teams are formed but have neglected the team formation phase (Dasgupta, Scircle, and Hunsinger 2015).

Given the importance of teamwork, understanding if individuals have equal opportunities to join a team is crucial to guarantee equality in career progress (Jones 2021). Women are one of the categories that might encounter barriers in accessing teams. Unique to our study, we look at the team formation phase assessing if this phase is affected by gender bias. By doing so, we contribute to the broad debate on gender equality. We analyze gender bias in team formation, considering academia, an ideal context in which individuals team up voluntarily, selecting teammates based on their preferences and expectations. Differently from previous studies, we observe teams in which the composition results from a deliberate choice of individuals without any filtering or external assessment. Indeed, we consider a sample of observed teams of scientists who, between 2006 and 2010, applied for the European Collaborative Research initiative promoted by the ESF. Our study includes teams awarded and non-awarded with the grant. During the team formation phase, a PL is required to form her own team by asking PIs to join. To assess gender bias,

we looked at the measurable characteristics of the observed PIs—with the exclusion of gender—and identified for each of those PIs a potential PI with similar characteristics that might have joined the team but did not.

To obtain a reliable control group of potential PIs, we implemented a three-step procedure assuming that a PL identifies her teammates among former colleagues with whom she worked, colleagues of whom she knows the work, and scientists working in departments in research fields of interest for the PL's project. In the first step, we retrieved a pool of potential PIs, including the coauthors of the PL, scientists cited in the PL's work, and scientists affiliated with the departments from which the PL selected the PIs in our observed teams. In the second step, we restricted the pool to those scientists with the competencies to complete the PL's project. Finally, we extracted the potential PIs with the closest profile to the observed PIs. Specifically, we found for each observed PI a 'twin' similar in terms of publications, career length, and the dimension of the coauthorship networks. By comparing the proportion of female scientists in the observed PIs' group with the same proportion in the group of potential 'twin' PIs, we found a gender bias against females. To further detail this result, we explore teams' heterogeneity. We find that gender bias is less pronounced in teams led by women, junior scientists, and in recently formed teams. We also find that physics and engineering are gender-neutral, while life and medical sciences, and humanities are gender biased against women. High- and low-productive PLs show a gender bias against women, although the bias is less pronounced for high-productive PLs. We also find gender bias in teams led by PLs working in high and low-masculine countries; however, gender bias is slightly more pronounced for PLs working in high masculine countries. Finally, the project quality does not affect gender bias in PIs' selection. Neither awarded nor non-awarded teams are gender biased.

Our findings speak to the current call for gender parity coming from multiple actors. For example, the most recent G7 dashboard on gender claims, 'Gender equality is a fundamental prerequisite for resilient, inclusive democratic societies. Nevertheless, gender inequalities continue to persist in all areas of social and economic life. Closing these gaps is an important goal and priority for all G7 members' (OECD 2022). In the last decades, plans and projects aiming to foster gender equality have been implemented for science and technology in Europe (European Commission 2021; European Union 2022; GEECCO Project 2023). Over the period covered by our analysis, several actions to reduce the gender gap have been implemented at the European, national, and institutional levels (Müller, González, and Palmen 2010). The reduction of the gender gap we observed when looking at recently formed teams seems to witness the efficacy of those actions. However, further policy interventions can be implemented to accelerate the fulfillment of the gender parity goal. Among the possible policy interventions, Müller, González, and Palmen (2010) list gender quotas, support for women's scientific network development, increasing transparency in team formation, support for women's participation in science, and incentives to disseminate female role models, and the creation of university-based gender observatories. In our context of scientists teaming up to apply for funds, introducing mandatory quotas of female scientists in teams tackles the problem of gender bias in team formation directly. For instance, the European Union has recently introduced gender balance among the ranking criteria to screen projects

that obtained the same evaluation score (European Union 2022). Moreover, we expect policy interventions supporting women's scientific networks to reduce gender bias effectively. Indeed, enlarging female scientists' professional networks makes women more visible within the scientific community and more likely to be selected as PIs. Another possibility to avoid gender bias in team formation could be a policy intervention that calls for a transparent justification of the team composition choices. For instance, the PL might be asked to motivate why the PI's profile fits the team in order to avoid applying any nonscientific criteria that might lead to discrimination. Our findings show that female PLs leading teams are not gender biased. According to this result, one possible way to stimulate gender parity might be incentivizing women to act as PLs. To achieve that goal, the ESF might establish gender quotas when selecting PLs. More women acting as PLs might be seen as role models by other women creating a reinforcing loop and increasing the share of women applying to ESF programs. Finally, the ESF might introduce a requirement to establish a gender observatory for the universities that intend to participate in their funding programs. The presence of gender observatories would create awareness of the gender bias problem (if any) in all the universities that intend to participate in the funding agency's calls for projects.

To conclude, our study has three potential limitations. One possible limitation is that we draw part of our control group of potential PIs, including former coauthors of the PL and scientists cited by the PL. PLs who are not gender-neutral might introduce a bias when choosing both coauthors and in citing authors' work. However, if this is the case, the gender bias against female scientists in team formation identified by our analysis would be underestimated. Indeed, our results show that female PIs are less likely to be observed in teams than in a control group where women are already potentially under-represented. A second limitation might be that, although our approach attempts to identify potential PIs for project teams based on observables, the actual formation of the project team results from both observed and unobserved factors. Our analysis assumes that the PL identifies PIs with the scientific competencies and technical skills needed to complete the project by retrieving that information from bibliometric data, i.e. coauthorships and citations. However, bibliometric data do not inform us about all PIs' characteristics, while PLs might acquire additional information on PIs from other sources, such as discussions with colleagues. For instance, PIs' working style cannot be inferred from bibliometric data and PLs might prefer to invite PIs with the reputation of being hard-working and able to work in teams. Moreover, the PL might have information about PIs' access to specific equipment or material resources useful for the project. Finally, even if we could observe PIs' characteristics using other data sources, we cannot observe PLs' preferences. For instance, the PL might have unobserved preferences for inviting PIs similar to her regarding cultural and educational background, age, and language. All these unobservable characteristics might bias the results of our analysis. Indeed, we assume that our selection on observables includes all the relevant characteristics that lead the PL to invite PIs to join the team. A final limitation of our study is that, although we empirically document gender inequalities, we can only speculate on possible mechanisms generating them. In other words, as a large part of the gender studies using secondary data, we cannot answer questions like 'Why shall a PL prefer male colleagues to female colleagues?'. We can only propose plausible explanations in our context.

For instance, the PLs' bias against teaming up with female PIs might be due to the females' stigma of having less probability of being funded. However, if this is the explanation, it remains unexplained why female scientists are discriminated by male PLs, but not by female PLs, as shown by our results. Further qualitative studies collecting interview data from PLs and PIs can well complement our quantitative findings.

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Appendix 1: Distribution of PIs and PLs by country

Table A.1. The number of PIs and PLs by the country of affiliation.

Country	Number of PIs	Number of PLs
Germany	248	73
Netherlands	114	46
Switzerland	94	21
Austria	89	21
UK	84	22
France	76	19
Italy	59	26
Sweden	57	16
Norway	53	14
Belgium	51	14
Poland	42	2
Finland	41	11

(continued)

Table A.1. (Continued)

Country	Number of PIs	Number of PLs
Spain	34	3
Czech Republic	28	4
USA	28	4
Portugal	22	3
Denmark	21	7
Turkey	21	1
Romania	15	2
Slovenia	13	2
Hungary	12	4
Ireland	10	2
Israel	10	2
Croatia	8	3
Slovak Republic	8	2
Estonia	7	2
Kenya	3	
Lithuania	3	1
Luxembourg	3	1
Bulgaria	1	
Cyprus	1	1
Greece	1	
Japan	1	
South Africa	1	
Ukraine	1	
Total	1,260	329

Appendix 2:

Potential PIs' selection validity test

Crucial to our analysis is the selection of the potential PIs. To identify the potential PIs, we assumed that the PL draws from three pools of researchers: (1) researchers with whom the PL collaborated in the past, (2) reputed colleagues she knows the work, and (3) researchers working in departments having competencies functional to complete the project proposed to the ESF. Although these seem reasonable ways for a PL to select PIs, there might be others. In this appendix, to validate our assumption on the potential sources of talent, we assess how many of the observed PIs have been drawn by the PL from the three pools. [Table B.1](#) shows the results of this validation exercise. Specifically, 53 per cent of the PIs selected in the application teams are drawn from one of the three pools. This high value shows that PL coauthors, referenced authors, and PIs' departmental colleagues are pools of researchers relevant to the PL's team selection process.

Table B.1. The average number of observed PIs per project application drawn from the three pools and the share of researchers drawn from each pool.

	Pool 1: PL's coauthors	Pool 2: PL's cited authors	Pool 3: colleagues of the observed PIs	Average number of observed PIs per project
Observed PIs	33%	20%	78%	3.83 overall, 2.04 (53%) in the three pools